

Full Name: SOLUTIONS

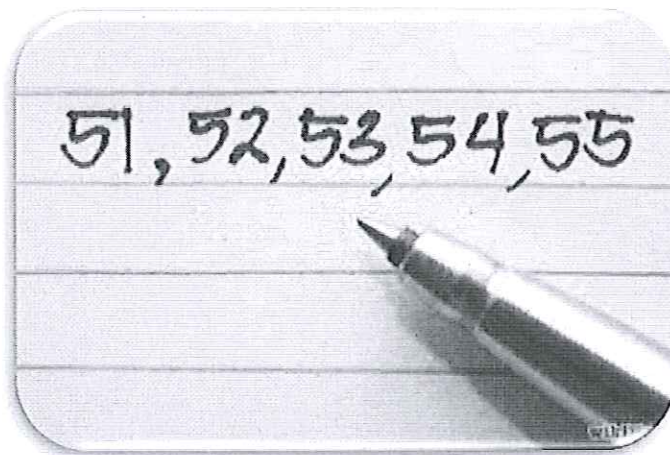


Ellenbrook
CHRISTIAN COLLEGE

MATHEMATICS
YEAR 8

Investigation 4 – Consecutive Numbers

Semester 2 2014



Information for Students

Working time:	50 minutes
Marks available:	42 marks

1. (14 marks)

The number 6 can be written as the sum of 3 consecutive whole numbers.

$$6 = 1 + 2 + 3$$

Consecutive numbers follow one after another. They are next to each other in order.

Show the possible ways you can write the numbers 2 to 16 as sums of consecutive numbers?

Note: This will not be possible for all numbers.

2	Not possible	✓
3	1 + 2	✓
4	Not possible	✓
5	2 + 3	✓
6	1 + 2 + 3	✓
7	3 + 4	✓
8	Not possible	✓
9*	4 + 5 or 2 + 3 + 4	✓
10	1 + 2 + 3 + 4	✓
11	5 + 6	✓
12	3 + 4 + 5	✓
13	6 + 7	✓
14	2 + 3 + 4 + 5	✓
15*	1 + 2 + 3 + 4 + 5 or 4 + 5 + 6 or 7 + 8	✓
16	Not possible	✓

* For 9 and 15 only one answer is needed

2. (3 marks)

a. Which of these numbers could **not** be written as the sum of consecutive whole numbers? [2]

2, 4, 8, 16

① two or three numbers

② Four numbers

b. Is there a pattern in these numbers? [1]

powers of 2 ✓

3. (4 marks)

a. Pick 6 pairs of consecutive whole numbers and make a list of their sums: [3]

	Numbers	Sum
	Example: 2 + 3	5
	Example 2: 3 + 4	7
1	4 + 5	9
2	5 + 6	11
3	6 + 7	13
4	7 + 8	15
5	8 + 9	17
6	9 + 10	19

1/2 each line
any valid
pair

b. Is there anything in common to these sums? [1]

All odd numbers ✓

4. (6 marks)

a. Pick 6 sets of 3 consecutive numbers and make a list of their sums: [3]

	Numbers	Sum
	Example: 1 + 2 + 3	6
1	2 + 3 + 4	9
2	3 + 4 + 5	12
3	4 + 5 + 6	15
4	5 + 6 + 7	18
5	6 + 7 + 8	21
6	7 + 8 + 9	24

1/2 each line
any valid
set

b. What type of numbers are the sums?

[1]

multiples of 3 ✓

c. What relationship exists between the middle number and the sum?

[1]

3 times middle number is the sum

✓ valid statement

d. How could you calculate $107+108+109$ without adding the numbers?

[1]

$$3 \times 108 = 324$$

✓

5. (5 marks)

a. Pick 6 sets of 4 consecutive numbers and list their sums and half their sums:

[3]

	Numbers	Sum	Sum / 2
	Example: $2 + 3 + 4 + 5$	14	7
1	$3 + 4 + 5 + 6$	18	9
2	$4 + 5 + 6 + 7$	22	11
3	$5 + 6 + 7 + 8$	26	13
4	$6 + 7 + 8 + 9$	30	15
5	$7 + 8 + 9 + 10$	34	17
6	$8 + 9 + 10 + 11$	38	19

*✓ each line
orig valid
set*

b. What do you notice about these results?

[1]

sum is even

sum/2 is odd

✓

c. What is the connection between the sum and the middle 2 numbers?

[1]

*sum of middle two numbers
is equal to the sum
divided by two*

✓ valid statement

6. (5 marks)

a. Pick 6 sets of 5 consecutive whole numbers and list their sums:

[3]

	Numbers	Sum
1	$1+2+3+4+5$	15
2	$2+3+4+5+6$	20
3	$3+4+5+6+7$	25
4	$4+5+6+7+8$	30
5	$5+6+7+8+9$	35
6	$6+7+8+9+10$	40

$\frac{1}{2}$ each line
any valid set.

b. What pattern exists between the numbers and their sums?

[1]

middle number times 5 ✓

c. How can you calculate $41+42+43+44+45$ without adding the numbers?

[1]

$$43 \times 5 = 215 \quad \checkmark$$

7. (5 marks)

a. Starting with $3+4$, repeatedly add a consecutive integer to both ends. Repeat 5 times.

[4]

The first repeat is done for you. Hint: You can use a zero and negative numbers.

$$3 + 4$$

$$2 + 3 + 4 + 5$$

$$1 + 2 + 3 + 4 + 5 + 6 \quad \checkmark$$

$$0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 \quad \checkmark$$

$$-1 + 0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 \quad \checkmark$$

$$-2 + -1 + 0 + 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 \quad \checkmark$$

b. Explain why the sum is always a multiple of 7.

[1]

7, 14, 21, 28, 35, 42

Each line the one before and one after add to 7
hence multiple of 7. ✓

End of Investigation